

- -Helix: Geometric Unification of Fundamental Constants

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Abstract

We present a geometric reinterpretation of three of physics' most fundamental numbers — the circle constant π , the fine-structure constant α , and particle mass m — as three orthogonal projections of a single geometric object: a helical toroidal vortex with the Compton wavelength as its characteristic scale. The identities $\alpha = r_e/\lambda$ and $m = \omega$ (in Planck units) follow directly from the definitions of the classical electron radius, Compton wavelength, and Planck units; their geometric significance emerges when mapped onto Hestenes' 2008/2019 helical Zitterbewegung model, in which the electron is a lightlike particle executing helical motion with diameter λ_c and radial thickness proportional to α . We trace the historical reification of π from Attic numeral $\Pi = 5$ to the modern idealized circle constant. The central thesis is that the appearance of independent “fundamental constants” is an artifact of projecting helical geometry onto lower-dimensional linear and circular subspaces. We note that this interpretation is currently limited to the electron; extension to other particles remains an open problem. We propose experimental signatures and identify falsification conditions.

Keywords: fine-structure constant, pi, Zitterbewegung, Hestenes, geometric unification, Compton vortex, reification, fundamental constants

1. Introduction — The Three Constants Problem

Physics rests on a set of numbers we call “fundamental constants.” Among them, three stand out for their ubiquity and elusiveness:

- $\pi \approx 3.14159 \dots$ — appears in formulas from Coulomb’s law to the Heisenberg uncertainty principle, from Gaussian integrals to Fourier transforms. It is treated as a pure number, a brute fact of mathematics.
- $\alpha \approx 1/137.036$ — the fine-structure constant, governing the strength of electromagnetic interactions. Feynman called it “one of the greatest damn mysteries of physics: a magic number that comes to us with no understanding by humans” (Feynman 1985). It is treated as a coupling strength, a measure of how strongly charged particles “couple” to the electromagnetic field.
- **Mass (m)** — the inertia of particles, measured in kilograms. It is treated as a property of “stuff,” an intrinsic attribute of material objects.

This paper argues that all three of these “constants” are the same thing, viewed from different angles. Specifically:

1. **α is a length ratio:** $\alpha = r_e/\lambda$, the classical electron radius divided by the reduced Compton wavelength. It is not a coupling strength but a geometric proportion — the ratio of the electromagnetic self-energy scale to the quantum oscillation scale.
2. **m is a frequency:** From $E = \hbar\omega$ and $E = mc^2$, we have $m = (\hbar/c^2)\omega$. In Planck units ($\hbar = c = G = 1$), this collapses to $m = \omega$. Mass IS angular frequency — not metaphorically, but algebraically.
3. **π is a projection artifact:** π appears where circular or spherical symmetry is hidden in the physics. In a helical geometry, the circle is a projection onto the transverse plane. What we call “ π ” is the residue of a helix collapsed to two dimensions.

These three insights converge on a single geometric object: the **Helical Compton Vortex** — a toroidal structure with Compton wavelength as its diameter, Compton frequency as its oscillation rate, α as its radial thickness, and π as its transverse projection. The electron, in this picture, is not a point particle with mysterious properties — it is a stable topological excitation of the electromagnetic vacuum, a helical standing wave whose geometry encodes all the numbers we have mistaken for independent constants.

2. Historical Foundations

2.1 The Dual Identity of π

The symbol π carries a double identity that has been almost entirely forgotten.

The Attic Π : $\Pi = 5$. In the Attic acrophonic numeral system, used in Athens from approximately the 7th to the 3rd century BCE, numbers were represented by the first letters of their Greek names (Wikipedia contributors 2026a). Π (pi) stood for π (pente), meaning “five.” The Attic numeral for 5 was written as Π or, in its decorative form, as Π (a pi with a decorative right stroke). This was not a variable or a constant — it was a cardinal number with a fixed, unambiguous value.

The full Attic system was:

Symbol	Value	Greek Word	Meaning
I	1	—	(tally mark)
Π ()	5	[pente]	five
Δ ()	10	[deka]	ten
H ()	100	[hekaton]	hundred
X ()	1000	[khilioi]	thousand
M ()	10000	[myrioi]	ten thousand

Composite numerals were formed by combining symbols: 50 was written as Δ inside Π (), and 500 as H inside Π (). The system was additive and multiplicative, elegant in its simplicity.

The Modern π : the circle constant. In 1706, the Welsh mathematician William Jones published *Synopsis Palmariorum Matheseos* (A Synopsis of the Palm-Bearers of Mathematics), in which he proposed using the Greek letter π to represent the ratio of a circle’s circumference to its diameter (Jones 1706). Jones chose π as an abbreviation of π (periphēria, “periphery” or “circumference”). This was the first documented use of π specifically for the circle constant (Wikipedia contributors 2026d, 2026e; MacTutor History of Mathematics, n.d.).

Before Jones, there was no universal symbol for the circle constant. William Oughtred (1647) used the notation π/δ to denote the ratio of periphery to diameter for *specific* circles; Isaac Barrow used π as a variable for perimeter in general. Archimedes had bounded the value between $3\frac{10}{71}$ and $3\frac{1}{7}$; Zu Chongzhi had computed $355/113$. But no one had given it a name. Jones did — and Leonhard Euler, beginning in 1736, adopted and standardized the notation, making π the universal symbol we know today.

The collision. The same letter Π/π thus carries two irreconcilable meanings: the cardinal number 5 (Attic) and the transcendental circle constant 3.14159... (modern). Both are valid within their systems, but the transition from one to the other tracks a deeper shift — from a worldview in which numbers were *counts of things* to one in which numbers were *abstract ideal forms*.

2.2 Pre-1706 Notations for the Circle Constant

Before Jones, the ratio of circumference to diameter was described in words, fractions, and ad-hoc symbols. A partial catalog:

Period	Culture/Person	Notation
~250 BCE	Archimedes	$3\frac{10}{71} < \pi < 3\frac{1}{7}$ (verbal)
~480 CE	Zu Chongzhi	$355/113$ (, “precise ratio”)
1647	Oughtred	π/δ for specific circles
~1660s	Barrow	π as variable for perimeter length
1706	Jones	π as universal circle constant

The lack of a universal symbol reflected the pre-modern understanding of the ratio: it was not a “number” in the modern sense but a *geometric proportion* — a relationship between two measurable quantities. The reification of π into a transcendent real number is a modern development.

2.3 Sommerfeld and the Fine-Structure Constant

Arnold Sommerfeld introduced α in 1916 to explain the fine structure of atomic spectral lines. He defined it as:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137}$$

The appearance of the velocity ratio $v_1/c = \alpha$ in the Bohr model (where $v_1 = e^2/(4\pi\epsilon_0\hbar)$ is the electron velocity in the first Bohr orbit) gave α its interpretation as a “coupling constant” — the strength of the electromagnetic interaction relative to the speed of light (Wikipedia contributors 2026c).

We will show that this interpretation, while historically productive, obscures a simpler geometric truth: α is a length ratio.

2.4 Planck and Natural Units

In 1899, Max Planck proposed a system of “natural units” based on the fundamental constants $\hbar$, c , and G (Planck 1899). In these units:

$$\hbar = c = G = 1$$

Under this normalization, many physical quantities collapse to dimensionless numbers. Most strikingly, mass becomes numerically equal to angular frequency:

$m = \omega$. This identity, implicit in Planck’s original formulation, is the key to understanding mass not as “stuff” but as a rate of phase accumulation.

3. Mathematical Foundations — Exact Identities

3.1 The α - r_e - λ Identity

The fine-structure constant admits an exact geometric interpretation that is mathematically trivial yet physically profound:

$$\alpha = \frac{r_e}{\lambda_e}$$

Derivation. We begin with three standard definitions from electrodynamics and quantum mechanics:

The fine-structure constant:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$$

The classical electron radius (the radius at which electrostatic self-energy equals rest mass energy):

$$r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2}$$

The reduced Compton wavelength (the scale at which quantum field effects become dominant):

$$\lambda_e = \frac{\hbar}{m_e c}$$

Dividing r_e by λ_e :

$$\frac{r_e}{\lambda_e} = \frac{e^2/(4\pi\epsilon_0 m_e c^2)}{\hbar/(m_e c)} = \frac{e^2}{4\pi\epsilon_0 m_e c^2} \cdot \frac{m_e c}{\hbar} = \frac{e^2}{4\pi\epsilon_0 \hbar c} = \alpha$$

This is an exact algebraic identity — no approximations, no model dependence. The fine-structure constant *is* the ratio of the classical electron radius to the reduced Compton wavelength.

3.2 The Full Scaling Chain

The identity $\alpha = r_e/\lambda$ implies a complete scaling chain linking all three fundamental electron length scales (Wikipedia contributors 2026b):

$$r_e = \alpha\lambda_e = \alpha^2 a_0$$

$$\lambda_e = \alpha a_0$$

$$a_0 = \frac{r_e}{\alpha^2}$$

Where $a_0 = 4\pi\epsilon_0\hbar^2/(m_e e^2)$ is the Bohr radius. All three scales — classical, Compton, and atomic — are locked together by powers of α :

Scale	Value (SI)	In terms of a_0	In terms of λ
Classical radius	2.818×10^{-15} m	$\alpha^2 a_0$	$\alpha\lambda$
r_e Reduced Comp- ton λ	3.862×10^{-13} m	αa_0	λ
Bohr radius	5.292×10^{-11} m	a_0	λ/α
a_0			

This hierarchy — spanning four orders of magnitude — is entirely encoded in powers of $\alpha \approx 1/137$. The “coupling constant” is, in fact, a geometric scaling factor.

3.3 Equivalent Forms of α as Length Ratios

Using the full Compton wavelength $\lambda_e = h/(m_e c) = 2\pi\lambda$:

$$\alpha = \frac{2\pi r_e}{\lambda_e} = \frac{\lambda_e}{a_0} = \sqrt{\frac{r_e}{a_0}}$$

All are exact. The fine-structure constant is a pure geometric proportion — a ratio of lengths — not a dynamically determined “coupling strength.”

3.4 CODATA 2022 Values with Uncertainties

Quantity	Symbol	Value	Uncertainty	Units
Inverse fine-structure	α^{-1}	137.03599917	± 0.000000021	dimensionless
Classical electron radius	r_e	$2.8179403205 \times 10^{-15}$	$\pm 1.3 \times 10^{-23}$	m
Reduced Compton wavelength	λ_e	$3.8615926764 \times 10^{-13}$	$\pm 2.5 \times 10^{-21}$	m
Compton wavelength	λ_e	$2.4263102389 \times 10^{-12}$	$\pm 1.6 \times 10^{-20}$	m
Bohr radius	a_0	$5.29177210544 \times 10^{-11}$	$\pm 8.2 \times 10^{-20}$	m

Verification: $r_e/\lambda_e = 2.81794 \times 10^{-15} / 3.86159 \times 10^{-13} = 0.0072973525643 = \alpha$.

Source: CODATA 2022 internationally recommended values (National Institute of Standards and Technology (NIST) 2022).

3.5 Geometric Proportion vs. Modern Constant

The distinction between “geometric proportion” and “fundamental constant” is more than semantic. In the Eudoxian-Euclidean tradition (*Elements* Book V, Definition 5), a proportion () is a relationship between four magnitudes: a is to b as c is to d (Euclid ~300 BCE). The magnitudes themselves have no intrinsic numerical value — they acquire meaning only through their ratios.

The modern concept of a “fundamental constant” inverts this: it treats α as a bare number (0.0072973525643...) that somehow encodes the strength of a force. But α is not a strength — it is a *shape*. It tells you the aspect ratio of the electron’s internal geometry: the electromagnetic self-energy radius is α times the quantum oscillation wavelength.

3.6 Dimensional Analysis

Both π and α are dimensionless — they are pure numbers. Mass, in SI units, carries dimensions of kilograms. But in Planck units, $m = \omega$ is also dimensionless. This means that all three “constants” are, in the appropriate natural units, pure geometric proportions.

4. Mass-Frequency Identity

4.1 The Chain of Equalities

Two of the most fundamental equations in physics are:

$$E = \hbar\omega \quad (\text{Planck-Einstein, 1900/1905})$$

$$E = mc^2 \quad (\text{Einstein, 1905})$$

Equating them:

$$\hbar\omega = mc^2$$

$$\boxed{m = \frac{\hbar}{c^2} \omega}$$

In SI units, mass and angular frequency differ by the conversion factor $\hbar/c^2 \approx 1.173 \times 10^{-51} \text{ kg} \cdot \text{s}$. This factor is the dimensional residue of our unit choices, not a physical constant.

4.2 Mass in Planck Units

In Planck's natural units, we set:

$$\hbar = c = G = 1$$

Under this normalization:

$$\boxed{m = \omega}$$

Mass IS angular frequency. The rest mass of a particle *is* the angular frequency of its internal oscillation. This is not an interpretation or an analogy — it follows directly from equating two experimentally verified relations and choosing natural units.

4.3 The Compton Relations

The Compton angular frequency:

$$\omega_C = \frac{mc^2}{\hbar}$$

In Planck units: $\omega_C = m$ (since $c = \hbar = 1$).

The reduced Compton wavelength:

$$\lambda = \frac{\hbar}{mc} = \frac{c}{\omega_C}$$

In Planck units: $\lambda = 1/m = 1/\omega_C$.

The Compton relations establish a complete equivalence between mass, frequency, and inverse length.

4.4 SI $\leftrightarrow$ Planck Unit Conversion Table

Quantity	SI Expression	SI Value (electron)	Planck Value
Mass m_e	—	9.109×10^{-31} kg	$m_P \times 4.185 \times 10^{-23}$
Compton frequency ω_C	$mc^2/\hbar$	7.763×10^{20} rad/s	$4.185 \times 10^{-23} t_P^{-1}$
Reduced Compton λ	$\hbar/(mc)$	3.862×10^{-13} m	$2.389 \times 10^{22} l_P$
ZB frequency ω_{ZB}	$2mc^2/\hbar$	1.553×10^{21} rad/s	$8.370 \times 10^{-23} t_P^{-1}$

Where $m_P \approx 2.176 \times 10^{-8}$ kg, $l_P \approx 1.616 \times 10^{-35}$ m, $t_P \approx 5.391 \times 10^{-44}$ s.

4.5 α as Phase Advance

Since $r_e = \alpha\lambda$ and $\lambda = c/\omega_C$:

$$\alpha = \frac{r_e}{\lambda} = r_e \cdot \frac{\omega_C}{c} = \frac{r_e/c}{1/\omega_C}$$

α is the ratio of (a) the light-crossing time of the classical electron radius to (b) the Compton period. In other words: α measures the phase advance of the Zitterbewegung oscillation over the electromagnetic self-energy scale.

5. The Dirac Electron and Zitterbewegung

5.1 The Dirac Equation

The Dirac equation (1928) for a free electron:

$$(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0$$

has a remarkable property: the velocity operator $\alpha = c\gamma^0\gamma$ has eigenvalues $\pm c$. The electron, in the Dirac picture, always moves at the speed of light (Wikipedia contributors 2026f).

5.2 The Zitterbewegung Solution

In the Heisenberg picture, the position operator of a free Dirac electron evolves as:

$$\mathbf{x}(t) = \mathbf{x}(0) + \frac{\mathbf{p}c^2}{H}t + \frac{i\hbar c}{2H} \left(\alpha(0) - \frac{\mathbf{p}c}{H} \right) e^{-2iHt/\hbar}$$

The first two terms describe classical linear motion. The third term is an oscillatory component — the *Zitterbewegung* (German: “trembling motion”) — with:

- **Angular frequency:** $\omega_{ZB} = 2mc^2/\hbar$ (twice the Compton frequency)
- **Amplitude:** $\hbar/(2mc) = \lambda/2$ (for a free electron at rest)

The electron is not at rest — it oscillates at 1.55×10^{21} Hz with an amplitude of 1.93×10^{-13} m, tracing out a path in spacetime.

5.3 The Velocity Operator $\pm c$

That the velocity operator has eigenvalues $\pm c$ is a direct consequence of the Dirac algebra. The electron is always “trying” to move at the speed of light, switching direction at twice the Compton frequency. The observed subluminal velocity is a time-averaged effect — the “rest mass” is the energy of this oscillatory motion.

5.4 Mainstream Interpretation

The mainstream interpretation, following Foldy and Wouthuysen (1950), treats the Zitterbewegung as an artifact of the single-particle Dirac equation — eliminated by a unitary transformation that separates positive- and negative-energy states. In quantum field theory, the ZB is interpreted as interference between positive- and negative-energy components, with no physical reality.

However, this elimination is not a disproof — it is a choice of representation. The Foldy-Wouthuysen transformation moves the oscillation into the wavefunction’s phase, where it becomes invisible but does not disappear.

5.5 Penrose’s Zigzag Picture

Roger Penrose, in *The Road to Reality* (2004), offered a vivid alternative: the electron as a massless particle zigzagging at the speed of light, alternately as a left-handed and right-handed Weyl fermion. The “rest mass” is the energy of

this confined light-speed motion. The zigzag frequency is the ZB frequency; the zigzag amplitude is the Compton wavelength (Penrose 2004).

Penrose’s picture, while heuristic, captures the essential physics: the electron is not a stationary object with “mass” as a property, but a confined light-speed process whose oscillation *is* the mass.

6. Hestenes’ Helical Zitterbewegung Model

David Hestenes, over two decades of work, developed the most complete geometric model of the electron based on the Dirac equation and Spacetime Algebra (STA) — a Clifford algebra formulation of relativistic physics that eliminates the need for Dirac matrices.

6.1 The 2008 Model: Lightlike Helical Particle

In *Zitterbewegung in Quantum Mechanics — a research program* (2008), Hestenes proposed a “self-contained dynamical model of the electron as a lightlike particle with helical zitterbewegung” (Hestenes 2008). Key results:

- The electron is a **lightlike** particle — its instantaneous velocity is always c . The subluminal observed velocity is an average over the helical path.
- The ZB is **helical**, not circular or random. The electron traces a cylindrical helix in spacetime.
- The model yields a specific prediction: the electron possesses an oscillating electric dipole moment (EDM) that should produce detectable resonance in electron channeling experiments through crystals.
- The spin is not an intrinsic “quantum” property but the orbital angular momentum of the lightlike helical motion.

Using STA, Hestenes reformulated the Dirac equation without matrices, making the geometric structure explicit. The electron’s wavefunction becomes a rotor — a geometric object describing rotation in spacetime — rather than an abstract complex-valued spinor.

6.2 The 2019 Model: Toroidal Vortex

In *Zitterbewegung structure in electrons and photons* (2019), Hestenes extended the model dramatically (Hestenes 2019). The central claims:

1. The Dirac equation is reinterpreted as a constitutive equation for singularities in the electromagnetic vacuum. The electron is not a “particle” subject to the Dirac equation — the Dirac equation *describes* the structure of a stable topological defect in the vacuum.
2. The electron is a point singularity on a lightlike toroidal vortex. The vortex has a well-defined geometry:

- **Diameter:** the Compton wavelength $\lambda_c = h/(m_e c)$
 - **Thickness:** proportional to the electron's anomalous magnetic moment, or equivalently to $\alpha \cdot \lambda_c/(2\pi)$
 - **Circulation:** lightlike — the vortex core circulates at c
3. The photon is modeled as an electron-positron pair trapped in a vortex ring. This provides a unified description of electrons and photons as different configurations of the same fundamental object — a lightlike circulation in the electromagnetic vacuum.
 4. All elementary particles may be composed of similar vortex structures. The model suggests a research program in which the “particle zoo” is reduced to topology.

6.3 Key Geometric Parameters

Extracted from Hestenes' 2008 and 2019 papers:

Parameter	Value	Interpretation
Vortex diameter	$\lambda_c =$ $h/(m_e c) =$ 2.426×10^{-12} m	Compton wavelength
Circulation speed	c	Lightlike vortex core
Circulation frequency	$\omega_{ZB}/2\pi =$ 2.47×10^{20} Hz	ZB frequency / 2π
Radial thickness	$\propto \alpha \cdot \lambda_c/(2\pi)$	Set by anomalous magnetic moment
π appears as	Transverse circumference factor	Helix $\rightarrow$ circle projection
α appears as	Radial-to-axial aspect ratio	Vortex thickness / diameter

6.4 Experimental Predictions

Hestenes' model makes a specific, falsifiable prediction: an electron moving through a crystal lattice should exhibit a resonant interaction when the channeling period matches the ZB period. This would manifest as an oscillating electric dipole moment detectable through radiation emission or energy loss spectra.

The resonance condition occurs when:

$$d/v = \pi \hbar / (mc^2)$$

where d is the crystal plane spacing and v is the electron velocity. This has not yet been experimentally confirmed or refuted, making it a critical test of the model.

6.5 Relation to Our Thesis

Hestenes’ model provides the physical mechanism underlying the mathematical identities we have established:

- $\alpha = r_e/\lambda$ maps to the vortex’s radial thickness relative to its diameter
- $m = \omega$ maps to the vortex’s circulation frequency
- π appears as the factor relating the helical path’s transverse circumference to the vortex diameter

The three “constants” are not independent — they are different measurements of the same vortex, taken along different axes.

7. Extended Electron Models

7.1 Consa’s Helical Solenoid Model (2018)

Oliver Consa extended Hestenes’ model into a “Helical Solenoid Model of the Electron” (Consa 2018). Key contributions:

- The electron is modeled as a helical solenoid — a current loop that also advances along its axis, producing both electric and magnetic moments
- The model derives the electron’s g -factor (≈ 2) directly from the helical geometry, without QFT loop corrections
- The anomalous magnetic moment $(g-2)/2$ is attributed to the solenoid’s pitch angle, which is proportional to $\alpha/2\pi$

In a related paper, Consa (2017) derived the g -factor explicitly from the helical geometry, obtaining $g = 2$ as the zero-thickness limit and $(g-2)/2 \approx \alpha/(2\pi)$ as the first-order thickness correction (Consa 2017). This connects α directly to the electron’s magnetic properties through geometry, not perturbation theory.

7.2 Rodrigues et al.: ZB and Electron Structure (1993)

Rodrigues, Vaz, Recami, and Salesi developed one of the earliest geometric models connecting Zitterbewegung to electron structure (Rodrigues et al. 1993). Using Clifford algebra, they showed:

- The electron’s spin can be derived as the orbital angular momentum of the ZB motion
- The ZB is not an artifact but a real physical oscillation
- The model naturally yields the de Broglie wavelength as a consequence of the ZB frequency combined with translational motion

7.3 Williamson & van der Mark: Toroidal Photon Topology (1997)

Williamson and van der Mark proposed that the electron *is* a self-confined photon on a toroidal path (Williamson and Mark 1997). In their model:

- A photon of energy mc^2 circulates on a closed toroidal path of circumference λ_c (the Compton wavelength)
- The self-interference of the circulating photon produces a standing wave — the electron’s rest frame wavefunction
- The topology (toroidal, single-loop) determines the spin- $\frac{1}{2}$ and charge quantum numbers
- The fine-structure constant α emerges as the ratio of the electromagnetic coupling energy to the photon’s confinement energy

7.4 Comparison Table

Model	Year	Geometry	ZB Real?	α Meaning	g -factor
Dirac (original)	1928	Point particle	Yes (osc.)	Coupling constant	2 (predicted)
Foldy- Wouthuysen	1950	Point particle	Eliminated	Coupling constant	2
Penrose zigzag	2004	Zigzag at c	Yes (heuristic)	—	—
Hestenes helix	2008	Lightlike helix	Yes (physical)	Pitch angle	—
Hestenes vortex	2019	Toroidal vortex	Yes (circu- lation)	Thickness/diam.	—
Consa solenoid	2018	Helical solenoid	Yes (current)	Pitch factor	$2 + \alpha/\pi$
Rodrigues et al.	1993	Clifford ZB	Yes (orbital)	—	From ZB
Williamson et al.	1997	Toroidal photon	Yes (circu- lation)	Coupling ratio	—
This work	2026	Helical Compton Vortex	Yes (def.)	$r_e/\lambda =$ aspect ratio	Geometric

8. The Unified Geometric Object

8.1 Definition: The Helical Compton Vortex

We propose that the electron is a **Helical Compton Vortex** — a stable, localized circulation in the electromagnetic vacuum with the following parameters:

Property	Value	Expression
Diameter	$\lambda_c = 2.426 \times 10^{-12} \text{ m}$	$h/(m_e c)$
Radius	$\lambda = 3.862 \times 10^{-13} \text{ m}$	$\hbar/(m_e c)$
Circulation speed	c	Lightlike
Angular frequency	$\omega_C = 7.763 \times 10^{20} \text{ rad/s}$	$m_e c^2/\hbar$
Transverse circumference	$\pi \lambda_c$	$\pi \times \text{diameter}$
Radial thickness	$(\alpha/2\pi)\lambda_c \approx 2.81 \times 10^{-15} \text{ m}$	Classical electron radius
Helix pitch per turn	$\alpha \lambda_c$	$2\pi \times \text{classical radius}$

8.2 π as Transverse Projection

When a helix is projected onto a plane perpendicular to its axis, it traces a circle. The circumference-to-diameter ratio of this projected circle is π . Every appearance of π in fundamental physics can be traced to this projection: the helix becomes a circle, the 3D structure becomes 2D, and π appears as the residue of the lost dimension.

In the Helical Compton Vortex: - The **vortex** is helical/toroidal — a 3D structure - The **projection** onto the transverse plane is circular — a 2D shadow - π is the **shadow's** circumference-to-diameter ratio — a property of the projection, not the object

8.3 α as Radial Thickness

The fine-structure constant α is the ratio of the vortex's radial thickness (classical electron radius r_e) to its axial scale (reduced Compton wavelength λ):

$$\alpha = \frac{r_e}{\lambda} = \frac{\text{radial thickness}}{\text{vortex radius}} \approx \frac{1}{137}$$

This interprets α not as a “strength” but as an *aspect ratio* — a geometric shape parameter. The electron is “thin” ($\alpha \ll 1$) because its electromagnetic self-energy scale is much smaller than its quantum oscillation scale. A coupling constant of $1/137$ is, in this picture, an aspect ratio of 1:137.

8.4 Mass as Oscillation Frequency

The electron's rest mass is the angular frequency of its helical circulation, measured in Planck units:

$$m = \omega_C = \frac{m_e c^2}{\hbar}$$

The “inertia” of the electron is the rate at which its internal phase accumulates. To accelerate the electron is to change the phase accumulation rate — and the resistance to this change is what we call mass.

This resolves the conceptual puzzle of mass: it is not a primitive property of “stuff” but a dynamical quantity — the frequency of an internal oscillation. The Higgs mechanism may give this oscillation its rest frequency, but the fact that mass IS frequency is a logical consequence of $E = \hbar\omega$ and $E = mc^2$.

8.5 The Unified View

The three “fundamental constants” are three orthogonal projections of the same geometric object:

Helical Compton Vortex	$\left\{ \begin{array}{ll} \pi & \text{— transverse projection (circle constant)} \\ \alpha & \text{— radial-to-axial ratio (aspect ratio)} \\ m = \omega & \text{— circulation frequency (inertia)} \end{array} \right.$
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They are not independent. Given the electron’s Compton wavelength (which sets the scale), α determines the radial thickness, and ω determines the circulation rate. π is not an input — it emerges from the helical geometry as the transverse projection factor.

9. Reification of Constants — A Critique

9.1 What Is Reification?

Reification (from Latin *res*, “thing”) is the fallacy of treating an abstraction as if it were a concrete, independent entity. In physics, reification occurs when we mistake:

- A **ratio** for a **property** (α : from length ratio to “coupling strength”)
- A **projection** for an **object** (π : from geometric proportion to transcendental number)
- A **frequency** for a **substance** (mass: from oscillation rate to “amount of stuff”)

9.2 The Reification of π

π is treated as a number — 3.14159... — a transcendental constant of mathematics whose decimal expansion is known to trillions of digits. But this is a profound reification.

In its original geometric context, π is not a number at all — it is a *proportion*, a relationship between two measurable quantities: the circumference and diameter of a circle. The Greeks understood this: they classified magnitudes, not numbers. A length was a length, and the ratio of two lengths was not a “number” in our sense but a *logos* (logos) — a rational relationship.

The modern treatment of π as a real number — and a transcendental one at that — turns a geometric relationship into an ontological object. We ask “what IS π ?” as if it were a thing, rather than “what relationship does π encode?” The answer to the second question is simpler: any system with hidden circular symmetry.

9.3 The Reification of α

$\alpha \approx 1/137$ has been called “the most mysterious number in physics.” Feynman wrote:

It has been a mystery ever since it was discovered more than fifty years ago, and all good theoretical physicists put this number up on their wall and worry about it. ... Immediately you would like to know where this number for a coupling comes from: is it related to π or perhaps to the base of natural logarithms? Nobody knows. It’s one of the greatest damn mysteries of physics. (Feynman 1985)

But if $\alpha = r_e/\lambda$ is an exact algebraic identity, then α is not a “magic number” — it is the aspect ratio of the electron’s internal geometry. The mystery is not “why 1/137?” but “why does the electron have this particular aspect ratio?” This shifts the question from numerology to geometry — a tractable problem.

The coupling constant interpretation is the result of reification: we took a geometric proportion and treated it as a dynamical parameter governing interaction strength. The mathematics doesn’t distinguish these interpretations — $\alpha = e^2/(4\pi\epsilon_0\hbar c) = r_e/\lambda$ — but the ontology does.

9.4 The Reification of Mass

Mass is perhaps the most deeply reified concept in physics. We speak of “massive particles” as if mass were a substance they possess. The SI system has a dedicated base unit — the kilogram — for measuring this “quantity of stuff.”

But $m = \omega$ in Planck units tells us that mass is a frequency — a rate of phase accumulation. The electron’s “rest mass” of 9.109×10^{-31} kg is, in more fundamental terms, an oscillation at 7.763×10^{20} rad/s. We don’t perceive the oscillation because it is far too fast, and because our measuring instruments are calibrated to report mass in kilograms rather than frequency in hertz.

The practical utility of measuring mass in kilograms is not in dispute. The conceptual error is in treating kilograms as measuring a *different kind of thing*

from hertz. They measure the same thing — oscillation frequency — in different units.

9.5 Historical Parallel: Greek Proportion Theory

The ancient Greek distinction between number ($\alpha\rho\iota\theta\mu\omicron\varsigma$, arithmos) and magnitude ($\mu\epsilon\gamma\epsilon\theta\omicron\varsigma$, megethos) is instructive. Numbers were counts of discrete units; magnitudes were continuous quantities (lengths, areas, times). A proportion ($\alpha\iota\sigma\chi\eta$) related four magnitudes — $a : b :: c : d$ — without assigning numerical values to any of them.

The modern real number system collapsed this distinction: every magnitude is assigned a real number. This was a triumph of mathematical unification but a loss of conceptual clarity. When we say “ $\pi = 3.14159\dots$ ” we are treating a proportion as a number — reifying the shadow.

9.6 Linear/Circular Assumptions in Physics

Most of physics is built on linear algebra (vectors, Hilbert spaces) and circular harmonics (Fourier series, spherical harmonics). These are mathematically convenient but geometrically limiting: they assume that the fundamental objects are points (0D), lines (1D), circles (2D closed), or spheres (3D closed).

A helix is none of these. It is 3D, open, chiral, and fundamentally different from both lines and circles. If the electron is a helix, then:

- Vector formulations are projections onto a lower-dimensional subspace
- Fourier decompositions decompose the helix into circular components that aren’t individually physical
- π appears because the projection is circular, not because the object is

10. Catalog of π Appearances in Physics

If π is the transverse projection of helical geometry, every π in a physics formula should be traceable to a hidden circular or spherical symmetry.

10.1 Major π Appearances with Geometric Interpretation

Formula	π Role	Geometric Origin
Coulomb: $F = q_1 q_2 / (4\pi\epsilon_0 r^2)$	4π from Gauss flux through sphere	Spherical symmetry of point source
Uncertainty: $\Delta x \Delta p \geq \hbar/2 = h/(4\pi)$	2π from wave periodicity	Fourier duality of conjugate variables

Formula	π Role	Geometric Origin
Hydrogen energy: $E_n = -me^4/(8\varepsilon_0^2\hbar^2n^2)$	Hidden in $\alpha = e^2/(4\pi\varepsilon_0\hbar c)$	Circular Bohr orbits
Blackbody: $\sigma = 2\pi^5k^4/(15h^3c^2)$	$2\pi^5$ from angular integrals	Spherical integration
Schrödinger: $i\hbar\partial_t\psi = -(\hbar^2/2m)\nabla^2\psi$	π in $\hbar = h/(2\pi)$	Wave nature of matter
Dirac: $(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0$	π in $\hbar$	Relativistic wave equation
Einstein field: $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$	8π from spherical symmetry	Gauss-Bonnet in 4D
Schwinger: $a_e = \alpha/(2\pi)$	2π from angular integration	Loop Feynman integral

10.2 Classification Scheme

Category	Description	Examples
Circumferential	$C = 2\pi r$, area $= \pi r^2$	Coulomb's law, Gauss's law
Spherical	Surface area $= 4\pi r^2$, solid angle	Planck's law, blackbody
Fourier/Periodic	$e^{i2\pi x/\lambda}$, Fourier transform	Quantum mechanics, signals
Gaussian	$\int e^{-x^2} dx = \sqrt{\pi}$	Path integrals, statistics
Angular momentum	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$, eigenvalues	Atomic physics, spin

10.3 The Helical Interpretation

In the Helical Compton Vortex picture, all these π factors trace back to the same geometric source: the projection of the helical electron onto a transverse plane. The Coulomb 4π , the uncertainty 2π , the Schrödinger π in $\hbar$ — all are different facets of the same hidden circularity.

The electron's internal geometry is helical. When we measure it in a linear or circular basis, π appears as the residue of the projection. A truly helical formulation of quantum mechanics — perhaps in Hestenes' Spacetime Algebra — would make the π factors disappear, absorbed into the geometric structure of the rotor representation.

10.4 π -Free Formulations

Law	Contains π ?	Why?
$F = ma$	No	Linear — no hidden circle
$E = mc^2$	No	Linear — no hidden circle
$E = \hbar\omega$	Yes (in $\hbar$)	Wave periodicity
Coulomb's law	Yes (4π)	Spherical flux
Schrödinger eq.	Yes (in $\hbar$)	Wave nature
Einstein field eq.	Yes ($8\pi G$)	Spherical symmetry

The pattern is clear: π appears whenever the physics involves rotation, oscillation, or spherical symmetry — exactly the structures that emerge when a helix is projected onto lower-dimensional subspaces.

11. Implications and Predictions

11.1 If the Electron IS a Helical Compton Vortex

The model makes several testable predictions beyond Hestenes' channeling resonance:

1. **π -free formulation of QM.** A complete reformulation of quantum mechanics in Spacetime Algebra should eliminate all explicit π factors, with the geometric structure absorbing them into rotor representations.
2. **α from geometry.** The value of α should be derivable from the vortex's stability conditions — a balance between electromagnetic self-energy and quantum oscillation energy. This is a well-posed variational problem.
3. **Mass ratios.** If all particles are vortex configurations, their mass ratios should correspond to ratios of topological invariants (winding numbers, linking numbers).
4. **Universality of α .** The same aspect ratio should appear in any system where electromagnetic self-energy is balanced against quantum oscillation — not just electrons, but muons, tauons, and possibly hadronic systems.

11.2 Relationship to the Anomalous Magnetic Moment

The electron's anomalous magnetic moment $a_e = (g-2)/2 \approx 0.00116$ is currently computed to 12 significant figures in QED perturbation theory and matches experiment to similar precision.

In the vortex model, $a_e \approx \alpha/(2\pi) \approx 0.00116$ at leading order — the first-order thickness correction. The QED perturbative series:

$$a_e = \frac{\alpha}{2\pi} + C_2 \left(\frac{\alpha}{\pi}\right)^2 + C_3 \left(\frac{\alpha}{\pi}\right)^3 + \dots$$

may have a geometric reinterpretation: each term corresponds to a higher-order correction to the vortex shape (self-interaction of the helical current). The extraordinary precision of QED would then not contradict the geometric model — it would *validate* it.

11.3 Connection to the Electroweak Scale?

If the electron is a vortex in the electromagnetic vacuum, other particles may be vortices in other fields. The electroweak scale ($v \approx 246$ GeV) may represent the energy at which the vortex character becomes explicit — where the “particle” approximation breaks down and topology takes over.

The ratio $m_e/v \approx 3.7 \times 10^{-6}$ is vastly smaller than $\alpha \approx 1/137$. This hierarchy may have a topological explanation: the electron is a low-energy bound state (a vortex ring), while the W and Z bosons are excitations of the underlying field.

11.4 What Would Falsify the Model?

A scientific model must be falsifiable. The Helical Compton Vortex model would be falsified by:

1. **Experimental disconfirmation of Hestenes’ channeling resonance prediction** — if dedicated experiments find no ZB-related resonance after exhaustive search.
2. **A conclusive theoretical proof that ZB is unphysical** — not just eliminable by a unitary transformation, but provably non-existent.
3. **Discovery that α varies with energy scale in a way incompatible with geometric interpretation** — if α ’s running cannot be mapped to a vortex deformation.
4. **A competing model that explains the same facts with fewer assumptions** — Occam’s razor applies.

12. Conclusion

12.1 Summary of the Unification

We have shown that:

1. $\alpha = r_e/\lambda$ is an exact algebraic identity — the fine-structure constant is the ratio of the classical electron radius to the reduced Compton wavelength.

2. $m = \omega$ in Planck units is an identity following from $E = \hbar\omega$ and $E = mc^2$ — mass is angular frequency.
3. π is a projection artifact — the circle constant emerges when helical geometry is projected onto a transverse plane.
4. These three converge on a single geometric object — the Helical Compton Vortex — with Compton wavelength as diameter, α as aspect ratio, and ω_C as circulation frequency.

The three “fundamental constants” π , α , and m are not independent arbitrary numbers. They are three orthogonal projections of a single geometric structure: a helical toroidal vortex in the electromagnetic vacuum.

12.2 Open Questions

1. **What determines the absolute scale?** The Compton wavelength $\lambda_c = h/(m_e c)$ sets the scale, but h and c are fixed by definition in SI. In natural units, why $m_e \approx 4.2 \times 10^{-23} m_P$?
2. **Why $\alpha \approx 1/137$?** We have reinterpreted α as an aspect ratio, but the value of that ratio remains unexplained. Stability analysis of the vortex may yield it.
3. **What about the muon and tau?** Are heavier leptons vortices with different topological invariants? The mass ratios $m_\mu/m_e \approx 207$ and $m_\tau/m_e \approx 3477$ may encode topological information.
4. **Can the model be quantized?** The Helical Compton Vortex is a classical geometric model. Its quantization remains an open problem.
5. **Is there a deeper principle?** If all three “constants” are projections of one object, what principle determines the object’s properties?

12.3 Call for Experimental Tests

The most urgent next step is experimental: test Hestenes’ channeling resonance prediction. If the electron exhibits resonant behavior at the ZB frequency in crystal channeling experiments, the physical reality of the helical ZB is confirmed. If not, the model must be revised or abandoned.

The geometric unification presented here is elegant but provisional. It awaits the verdict of experiment.

Appendix A: Pre-1706 Notations for the Circle Constant

Date	Source	Notation	Description
~250 BCE	Archimedes	Verbal bounds	$3\frac{10}{71} < \pi < 3\frac{1}{7}$
~150 CE	Ptolemy	3; 8, 30 (sexagesimal)	$377/120 \approx 3.14167$
~480 CE	Zu Chongzhi	355/113 ()	Accurate to 6 decimal places
~1424	Al-Kashi	2 π to 16 dec. places	Sexagesimal and decimal places
1596	Ludolph van Ceulen	20 then 35 dec. places	“Ludolphine number”
1647	William Oughtred	π/δ	Periphery/diameter for specific circles
1655	John Wallis	Infinite product	Wallis product for $\pi/2$
~1660s	Isaac Barrow	π as variable	Perimeter length variable
1706	William Jones	π	Circle constant — first modern use

Appendix B: Attic Numeral System — Full Symbol Table

Symbol	Unicode	Value	Greek Word	Etymology
I	U+0399	1	—	Tally mark (iota)
II	U+03A0	5	(pente)	“five”
	U+10143	5	(decorative II)	Pi with right stroke
Δ	U+0394	10	(deka)	“ten”
	U+10144	50	—	Δ inside II
H	U+0397	100	(hekaton)	“hundred”
	U+10145	500	—	H inside II
X	U+03A7	1000	(khilioi)	“thousand”
	U+10146	5000	—	X inside II
M	U+039C	10000	(myrioi)	“ten thousand”
	U+10147	50000	—	M inside II

Appendix C: Dimensional Analysis Proofs

Theorem C.1: α , π , and m_P (in Planck units) are all dimensionless.

Proof of α : $\alpha = e^2/(4\pi\epsilon_0\hbar c)$. In SI: $[e^2] = \text{C}^2$, $[\epsilon_0] = \text{C}^2/(\text{N} \cdot \text{m}^2)$, $[\hbar] = \text{J} \cdot \text{s}$, $[c] = \text{m/s}$.

Computing: $[\alpha] = \text{C}^2/(\text{C}^2/(\text{N} \cdot \text{m}^2) \times \text{J} \cdot \text{s} \times \text{m/s}) = \text{C}^2 \cdot \text{N} \cdot \text{m}^2/(\text{C}^2 \cdot \text{J} \cdot \text{m}) = \text{N} \cdot \text{m}/\text{J} = 1$.

Proof for π : $\pi = C/d$ is the ratio of two lengths. Dimensionless by definition.

Proof for m_P : In SI, $[m] = \text{kg}$, $[\omega] = \text{s}^{-1}$. Conversion factor $\hbar/c^2$ has units $(\text{J} \cdot \text{s})/(\text{m}^2/\text{s}^2) = \text{kg} \cdot \text{s}$. So $m = (\hbar/c^2)\omega$ has consistent dimensions. In Planck units, $\hbar = c = 1$ eliminates the conversion, making m dimensionless.

Appendix D: STA Derivation (Placeholder)

Full Spacetime Algebra derivation deferred to a future version. In brief, Hestenes' STA reformulation of the Dirac equation eliminates complex numbers and Dirac matrices, replacing them with geometric (Clifford) algebra. The electron's wavefunction becomes a rotor $R(x)$ satisfying:

$$\nabla R = \frac{mc}{\hbar} R \gamma_0$$

The ZB emerges as the rotational component of R , and the velocity operator becomes a vector in spacetime rather than a 4×4 matrix. The helical path is a direct geometric consequence.

Appendix E: Complete Catalog of π in Physics Formulas

Formula	π count	Type
Coulomb: $F = q_1 q_2 / (4\pi \epsilon_0 r^2)$	4π	Spherical
Bohr magneton: $\mu_B = e\hbar/(2m_e)$	2π (in $\hbar$)	Angular
Uncertainty: $\Delta x \Delta p \geq \hbar/(4\pi)$	4π	Fourier
Hydrogen ground state: $\psi \propto e^{-r/a_0}$	0	—
de Broglie: $\lambda = h/p$	2π (in $\hbar$)	Periodic
Stefan-Boltzmann: $\sigma = 2\pi^5 k^4 / (15h^3 c^2)$	$2\pi^5$	Spherical
Einstein field: $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$	8π	Spherical
Schwinger: $a_e = \alpha/(2\pi)$	2π	Angular
Schrödinger: $-\hbar^2/(2m) \nabla^2 \psi$	2π (in $\hbar$)	Wave
Planck length: $l_P = \sqrt{\hbar G/c^3}$	Multiple	Definitional
Compton: $\lambda_c = h/(mc)$	0	Linear

Appendix F: SI $\leftrightarrow$ Planck Unit Conversion Tables

Quantity	Planck Unit	SI Value
Length	$l_P = \sqrt{\hbar G/c^3}$	1.616255×10^{-35} m
Time	$t_P = \sqrt{\hbar G/c^5}$	5.391247×10^{-44} s
Mass	$m_P = \sqrt{\hbar c/G}$	2.176434×10^{-8} kg
Energy	$E_P = \sqrt{\hbar c^5/G}$	1.956082×10^9 J
Temperature	$T_P = \sqrt{\hbar c^5/(Gk^2)}$	1.416784×10^{32} K

Electron Quantity	SI Value	Planck Value
Mass m_e	$9.1093837 \times 10^{-31}$ kg	$4.1854 \times 10^{-23} m_P$
Compton freq. ω_C	7.7634×10^{20} rad/s	$4.1854 \times 10^{-23} t_P^{-1}$
Reduced Compton λ_e	3.8616×10^{-13} m	$2.3893 \times 10^{22} l_P$
Classical radius r_e	2.8179×10^{-15} m	$1.7435 \times 10^{20} l_P$
Bohr radius a_0	5.2918×10^{-11} m	$3.2739 \times 10^{24} l_P$

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